magma freezes solid. Lava channels cannot go even 10 percent of the length of the long Venus channels before they solidify. Some planetary geologists think there must be a special thin, watery, inviscid lava generated on Venus. But this is a speculation supported by no other data, and a confession of our ignorance.

The thick atmosphere moves sluggishly; because it's so dense, though, it's very good at lifting and moving fine particles. There are wind streaks on Venus, largely emanating from impact craters, in which the prevailing winds have scoured piles of sand and dust and provided a sort of weather vane imprinted on the surface. Here and there we seem to see fields of sand dunes, and provinces where wind erosion has sculpted volcanic landforms. These aeolian processes take place in slow motion, as if at the bottom of the sea. The winds are feeble at the surface of Venus. It may take only a soft gust to raise a cloud of fine particles, but in that stifling inferno a gust is hard to come by.

There are many impact craters on Venus, but nothing like the number on the Moon or Mars. Craters smaller than a few kilometers across are oddly missing. The reason is understood: Small asteroids and comets are broken up on entry into the dense Venus atmosphere before they can hit the surface. The observed cutoff in crater size corresponds very well to the present density of the atmosphere of Venus. Certain irregular splotches seen on the *Magellan* images are thought to be the remains of impactors that broke up in the thick air before they could gouge out a crater.

Most of the impact craters are remarkably pristine and well preserved; only a few percent of them have been engulfed by subsequent lava flows. The surface of Venus as revealed by *Magellan* is very young. There are so few impact craters that everything older than about 500 million years* must have been eradicated—on a planet almost certainly 4.5 billion years old. There is only one plausible erosive agent adequate for what we see: vulcanism. All over the planet craters, mountains, and other geological features have been inundated by seas of lava that once welled up from the inside, flowed far, and froze.